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CIVIL WORKS DESIGN CRITERIA

UNIT 0000

GENERAL FACILITIES

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1 SCOPE

This document defines the minimum requirements for the design of concrete structures and defines the characteristics of materials to be used for T7 Onshore scope (i.e. CT, CLU, CSU, Flare, Utilities, CDF, OSBL infrastructure, Brownfield tie-ins, T7 Maintenance Workshop, New Passenger Terminal building, Workers Accommodation). This document is not applicable for the T7 Marine scope, which shall be based on T-14.111.800.

2 REFERENCE DOCUMENTS

For relevant Codes and Standards, the last valid edition (at the date of Contract award) shall be applied.

In particular the following documents shall be applied:

2.1 CODES AND STANDARDS

ACI 301M	Specification for Structural Concrete
ACI 318M	Building Code Requirements for Structural Concrete
ACI 336.2R	Suggested Analysis and Design Procedures for Combined Footing and Mats
ACI 350M	Code Requirements for Environmental Engineering Concrete Structures
ACI 224R	Control of Cracking in Concrete Structures
ACI 550.2R	Design Guide for Connections in Precast Jointed Systems
ACI 544.1R	Report on Fiber Reinforced Concrete
PCI-TR7	Precast/Prestressed Concrete Institute - Standard Design Practice
AASHTO	LRFD Bridge Design Specifications
ASTM C150/C150M	Standard Specification for Portland Cement

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ASTM A615/A615M	Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A1064/A1064M	Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM D6690	Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements
ASTM C881/C881M	Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C1107/C1107M	Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM C109/C109M	Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 50-mm Cube Specimens)
ASTM C579	Standard Test Methods for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing and Polymer Concretes
ASTM C595/C595M	Standard Specification for Blended Hydraulic Cements
ASTM C989/C989M	Standard Specification for Slag Cement for Use in Concrete and Mortars

2.1.1 **Shell General Functional Standard Specifications**

GFS IG.34.85.02-Gen.	Loading for Onshore LNG Facilities
GFS IG.34.85.20-Gen.	Foundation, Anchor Bolts and Grouting for Onshore LNG
GFS IG.34.85.21-Gen.	Design and Construction of Steel Structures for Onshore LNG Facilities
GFS IG.34.85.23-Gen.	Geotechnical Design of Shallow Foundations for

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Onshore LNG Facilities

GFS IG.34.85.25-Gen.	Heavy Machinery Foundations for Onshore LNG Facilities
GFS IG.34.85.31-Gen.	Design and Construction of Concrete Structures for Onshore LNG Facilities

Applicable technical deviations, if any, are presented in the Amendment sheets as listed in section 2.2.

2.1.2 Shell Design and Engineering Practice Specifications

DEP 34.17.10.32-Gen.	Design of Buildings – Onshore
DEP 34.19.20.11-Gen.	Passive Fire Protection for Onshore Facilities
DEP 34.13.20.31-Gen.	Roads, Paving, Surfacing, Cable Trenches, Slope Protection and Fencing

Applicable technical deviations, if any, are presented in Appendix A.

2.2 PROJECT DOCUMENTS

T-14.000.001-A	NLNG Train 7 Basis for Design - Midstream
T-14.111.002	Design Basis for Loads and Load Combinations
T-14.111.007	Specification For Anchor Bolts
T-14.111.103	Preloading Analyses Report Onshore Area
T-14.111.104	Basis of Foundation and Earthwork Design
T-14.111.600	Structural Steel Works Design Criteria
T-14.111.800	Design Basis & Civil Specification for Marine Facilities
T-14.112.000	Onshore Concrete Works General Notes
T-14.112.001	Anchor Bolts Standard Drawing

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T-14.112.006	Paving Typical Sections Drawing
T-14.112.079	Foundation for Pipe Support – Formwork & Reinforcement
T-14.111.008	Amendment to GFS IG.34.85.31-Gen. (February 2018) “Design and Construction of Concrete Structures for Onshore LNG Facilities”
T-14.111.009	Amendment to GFS IG.34.85.21-Gen (February 2018) “Design And Construction Of Steel Structures For Onshore”
T-14.111.010	Amendment to DEP 34.13.20.31-Gen. (February 2017) “Roads, Paving, Surfacing, Cable Trenches, Slope Protection and Fencing”
T-14.111.011	Amendment to GFS IG 34.85.02-Gen (February 2018) “Loading For Onshore LNG Facilities”
T-14.111.012	Amendment to GFS IG 34.85.20-Gen. (February 2018) “Foundation, Anchor Bolts and Grouting for Onshore LNG”
T-14.111.013	Amendment to GFS IG 34.85.23-Gen. (February 2018) “Geotechnical Design of Shallow Foundations for Onshore LNG Facilities”
T-14.111.018	Amendment to GFS IG.34.85.25-Gen (February 2018) “Heavy Machinery Foundations For Onshore LNG Facilities”

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2.3 UNIT OF MEASUREMENTS

The SI units will be used in design, in particular:

Length	m, mm
Mass	kg
Concentrated force, weight	N, kN
Linear distributed force	kN/m
Surface distributed force	kN/m ²
Specific weight	kN/m ³
Pressure, stress	N/mm ²
Temperature	°C

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3 MATERIALS AND COMPONENTS

3.1 CONCRETE

Concrete shall comply with the requirements of Chapter 19 of ACI318M, based on the exposure classes defined by the geotechnical documentation, and the compressive strength shall have a minimum strength after 28 days as follows:

Item	Design compressive strength of concrete f'_c [N/mm ²]
Slabs, area paving, roads, cover slabs for cable trenches, watertight concrete and machinery foundations	28
Structures and associated foundations, stationary equipment foundations with the exception of jetties	28
Seal slab or blinding concrete and miscellaneous non-structural concrete fill	14
Concrete envelopes for underground electrical ducts	14

The maximum Water-Cement Ratio (WCR) shall be as per the following:

Structure	WCR maximum
Typical Structures (except as Note 1)	0.45
Seal Slab or Blinding concrete and miscellaneous non-structural concrete fill	0.55
Liquid-retaining (hydraulic) structures and marine structures	0.40
Concrete deposited by tremie underwater or for prestressed concrete structures	0.44
Note 1: For mixtures containing pozzolan, the WCR is the water to (cement plus pozzolan) ratio	

Minimum cement content is 300 kg/m³.

Fibre reinforced concrete may be used only for non-structural applications, such as paving, in accordance with ACI 544.1R.

Permanent formwork for forming foundation structure (also known as Sacrificial

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Formwork) may be used.

The following target slumps at point of placement, determined as per ASTM C143/C143M, shall be adopted:

- V-ditches: 50 mm $\pm$ 25%
- All other castings: 100 mm $\pm$ 25%

Maximum size of coarse aggregate shall be 20 mm.

3.2 **CEMENT**

The entire Project shall be considered as Hydrocarbon Scope, with the exception of:

- Permanent Accommodation;
- Maintenance Workshop (MSW2);
- MOF (Material Offloading Facility) jetty 2 extension;
- New Passenger Terminal (NPT) & Former Aggregate Berth (FAB) Refurbishment Works.

For T7 Marine Scope facilities applications (i.e LNG Jetty 3, MOF jetty 2 extension, FAB Refurbishment Works) reference shall be made to Project Specification T-14.111.800.

Based on the chemical tests included in T-14.111.104, the water-soluble sulfate in soil is less than 0.10 % and the dissolved sulfate in water is less than 1500 ppm. Therefore, as per Table 19.3.1.1 of ACI 318M-19, Class S1 as Exposure category apply.

3.2.1 **Portland Cement**

For seal slab or blinding concrete, miscellaneous non-structural concrete fill, concrete envelopes for underground electrical ducts and all applications other than Hydrocarbon scope and Mass concreting (see sect. 3.2.2), the Cement shall be Type II according to ASTM C150/C150M (see table 19.3.2.1 of ACI 318M-19).

3.2.2 **Blended Cement**

For Hydrocarbon scope (i.e. all T7 process facilities and LNG jetty 3) and Mass concreting (see GFS IG.34.85.31-Gen "Design and Construction of Concrete Structures for Onshore LNG Facilities", section 5.9) cement shall be Blended Portland Blast Furnace Cement with minimum 65% blast furnace slag complying with the requirements of ASTM C595/C595M. Slag shall be compliant with ASTM

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C989/C989M.

Subject to approval by Company, blended cement with fly ash may be used in lieu of slag cement provided that Contractor demonstrates that the obtained concrete mix has the same durability, impermeability and resistance to prevailing environmental conditions as the concrete mix based on Blended Portland Blast Furnace Cement with minimum 65% blast furnace slag.

Fly ash shall be compliant with ASTM C618 Type C or Type F and the mix design shall be tested in accordance with ASTM C1012 to verify sulphate resistance.

3.3 **REINFORCING STEEL AND PRESTRESSING TENDONS**

Reinforcement bars shall conform to the following requirements:

- Steel type ASTM A615M Grade 60
- Minimum yield strength 420 N/mm²

Welded wire mesh shall conform to the following requirements:

- Type of welded wire ASTM A1064M
- Minimum yield strength 450 N/mm²

Prestressing reinforcement shall conform to the following:

- Strand ASTM A416M
- Wire ASTM A421M
- High strength bar ASTM A722M

3.4 **GROUT**

Grout shall be applied between the concrete foundation/pedestals and the baseplates of all steel structures, stationary equipment, rotating and reciprocating equipment and grout types shall be as follows:

Criteria considered	Non-shrink cement based grout	Epoxy grout
Driver horsepower	Less than 375 kW	More than or equal to 375 kW
Rotating equipment speeds	3600 RPM or less	More than 3600 RPM
Combined weight of (machine, driver, baseplate)	Up to 2268 kg	More than 2268 kg
All reciprocating machinery with driver	Less than 37 kW	More than 37 kW

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rating		
Other foundations	Non dynamic foundation	

Epoxy grout may be used at locations where exposure to chemical attack from regular operations is possible (for details refer to GFS IG.34.85.25-Gen. "Heavy Machinery Foundations for Onshore LNG Facilities" and Amendment sheet (T-14.111.018)).

3.4.1 **Non-Shrink Cement Based Grout**

Non-shrink grout shall be in accordance with ASTM C1107/C1107M and shall have minimum compressive strength $f'_c = 35$ MPa at 7 days and $f'_c = 40$ MPa at 28 days, tested in accordance with ASTM C109/C109M (for details refer to GFS IG.34.85.20-Gen "Foundation, Anchor Bolts and Grouting for Onshore LNG" and Amendment sheet (T-14.111.012)).

3.4.2 **Epoxy Grout**

The epoxy grout shall have minimum compressive strength at 7 days of 80 MPa, tested in accordance with ATSM C579, Method B (for details refer to GFS IG.34.85.20-Gen "Foundation, Anchor Bolts and Grouting for Onshore LNG" and Amendment sheet (T-14.111.012)).

3.5 **ANCHOR BOLTS AND EMBEDMENTS**

Anchor bolts materials shall be in accordance with the Project specification T-14.111.007 and standard drawing T-14.112.001.

Embedded steel material shall comply with ASTM A36/A36M and hot dip galvanized (except the anchorage designed based on embedment length) shall comply with GFS IG.34.85.21-Gen "Design and Construction of Steel Structures for Onshore LNG Facilities" (section 5.3) and Amendment sheet (T-14.111.009). Aluminum shall not be used for embedment materials.

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4 DESIGN CRITERIA

4.1 GENERAL

Reinforced concrete structures and foundations shall be designed in accordance with ACI 318M.

Bar spacing shall generally not exceed 300 mm for reinforcement placed near the concrete surface.

The minimum concrete cover, measured to the most exposed bar, tie or stirrup, shall be:

Cast-in-place non prestressed concrete members

- cast against and permanently in contact with ground 75 mm
- exposed to weather or in contact with ground 50 mm
- not exposed to weather or in contact with ground:
 - slabs, joists and walls 20 mm
 - beams, columns, pedestals and tension ties 40 mm

Cast-in-place prestressed concrete members

- cast against and permanently in contact with ground 75 mm
- exposed to weather or in contact with ground:
 - slabs, joists and walls 25 mm
 - all other 40 mm
- not exposed to weather or in contact with ground:
 - slabs, joists and walls 20 mm
 - beams, columns, pedestals and tension ties:
 - primary reinforcement 40 mm
 - stirrups, ties, spirals and hoops 25 mm

Precast non-prestressed or prestressed Concrete (manufactured under plant conditions)

- exposed to weather or in contact with ground:
 - walls 20 mm
 - all other:
 - for D20 mm bars or greater 40 mm
 - for D16 mm bars or smaller 30 mm
- not exposed to weather or in contact with ground:
 - slabs, joists and walls 20 mm
 - beams, columns, pedestals and tension ties:

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- primary reinforcement (greater of d_b and 16 mm but not more than 40 mm)
- stirrups, ties, spirals and hoops 10 mm

Precast Concrete (not manufactured under plant conditions)
The minimum covers for cast-in-place concrete shall apply.

In any case, for concrete exposed to hydrocarbons and prone to chemical attacks minimum cover shall be 50 mm.

Prestressed concrete shall only be allowed for bridges/culvert decks in roads and concrete LNG tanks.

Design of prestressed concrete shall be in accordance with ACI 318M and PCI TR-7;
anchorage device and local zone reinforcement shall meet the requirements of AASHTO 7, Articles 5.10.9.7.1 through 5.10.9.7.3.

Precast concrete members and connections shall be designed in accordance with ACI 318, ACI 550.2R and PCI TR-7.

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4.2 FOUNDATIONS

The minimum distance between the top surface of buried foundations and finished ground level will be 0.8 m, in order to avoid interference with underground works. *The design level of bottom of foundation shall be not less than 0.60 m from FGL.* This requirement is not applicable to foundations for pipe support (see T-14.112.079).

The bottom of the foundation shall be located such that it reaches adequate bearing stratum as per the recommendation of the geotechnical documentation.

The foundation (slab) thickness shall be the maximum of the following:

- 300 mm, increased in 50 mm increments;
- thickness required to develop the pedestal anchorage reinforcement;
- thickness required to resist the LRFD shear demand without shear reinforcement.

Foundations shall be provided with reinforcement to the top and bottom faces.

The top of foundation pedestal shall be a minimum 150 mm above the local finished paving level.

Minimum pedestal dimensions shall be 300 mm x 300 mm; pedestal shall be increased in 50 mm increments.

The pedestal height shall be at least equal to the anchor bolt embedded length.

Piers and pedestals shall be sized so that grouting can be extended 50 mm horizontally beyond the base/sole plate.

Unless specified by drawings, the grout thickness shall be within the range of 25 mm to 50 mm.

Minimum reinforcement limits, design and detailing of pedestals where the height to lateral dimension ratio is 3 or greater shall be designed as a column in accordance with ACI 318M and as follows:

- all longitudinal reinforcement shall be enclosed by lateral reinforcement that surrounds the vertical bars of the pedestal;
- minimum lateral reinforcement shall be provided in accordance with Section 10.6.2.2 of ACI 318M;
- lateral reinforcement shall be distributed within 100 mm of the top of the

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column or pedestal (lateral reinforcement shall consist of at least two D13 or three D10);

The minimum distance between the edge of base plate and the edge of concrete pedestal shall be as follows:

- 50 mm for steel structures, vertical vessels on legs, horizontal vessels and exchangers
- 75 mm for skirt supported vertical vessels and vibrating machines

Minimum clearance between centers of anchor bolts shall be not less than 6 diameters, except for horizontal equipment (vessels, exchangers), for which anchor bolts minimum spacing shall not be less than 4 diameters.

The minimum concrete edge distance (distance between anchor bolt axis and external face of concrete) and the minimum anchor bolt embedment length shall be as follows:

		Without sleeves		With sleeves
Concrete edge distance	Machine foundation	U	max (4da;155mm)	max (4da;155mm)+0.5(ds-da)
		T	max (6da;155mm)	max (6da;155mm)+0.5(ds-da)
	Structural foundation	U	max (4da;115mm)	max (4da;115mm)+0.5(ds-da)
		T	max (6da;115mm)	max (6da;115mm)+0.5(ds-da)
	All foundations	U	4da	4da+0.5(ds-da)
		T	6da	6da+0.5(ds-da)
Embedment length	All foundations	12da		for da≥25mm: max(12da;sleeve length+6da) for da<25mm: max(12da;sleeve length+155mm)
da = anchor bolt diameter; ds = anchor sleeve diameter; U = Un-torqued; T = Torqued				

When anchor bolts are placed in pockets, the minimum distance between the external surface of concrete and the nearest pocket side shall be 100 mm.

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Provision of nuts shall be as follows:

Equipment/Units	Double Nut
All structures, frames, racks, pipe supports, ancillary structures, field assembled units, equipment other than those noted below	
Modules and skids	
Guyed stack, derricks and communications towers	
Prefabricated equipment (boilers, heaters, etc.)	
Vertical vessels	X
Sphere	X
Stack, freestanding	X
Light machinery	X
Heavy machinery	X

A lean concrete layer of 50 mm minimum thickness shall be provided below all foundations.

Vertical vessel foundations shall be designed as rigid foundations (see T-14.111.000-SH2). Footings with face to face dimensions greater than 2100 mm shall be octagonal; the pedestal geometry shall be also octagonal, where face to face dimension ≥ 1200 mm.

In case Bolt Circle Diameter is greater than 4.0 m hollow pedestal shall be provided, unless otherwise noted in project drawings.

4.2.1 Shallow foundations

Soil stresses shall be calculated considering unfactored load combinations and will be compared with the allowable net bearing capacity defined in the geotechnical documentation.

For load combinations including variable actions such as wind loads or hydrostatic loads, the allowable bearing capacity can be increased of 25% for wind loads and of 20% for hydrostatic loads, if not otherwise indicated in the geotechnical documentation.

The stresses transmitted by the foundation to the ground shall be calculated as an equivalent uniform stress (Meyerhof Method) over a reduced contact area. The reduced area is the area of footing with respect to which the resultant force

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acts on the center of gravity.

Foundation shall be designed in order to limit the maximum total settlement to 50 mm and the maximum differential settlement, between adjacent foundations, to 25 mm.

The limit of 50 mm is not applicable to tank foundations and large-size foundations (side bigger than 10 m), for these items the value of the expected settlement shall be determined.

For large-size or raft foundations it is possible to deduct from the global final settlement the amount occurring during foundation construction, before equipment erection.

The following safety factors shall be considered in design:

- Overturning 2.0 (1.2 under blast/impact loads conditions)
- Sliding 1.5 (1.2 under blast/impact loads conditions)
- Uplift / buoyancy 1.1

The weight of soil above footings shall be considered for stability checks.

Overturning safety factor will be evaluated with respect to one edge for isolated foundations; for framed continuous structures, connected to the foundation by a moment resisting element, only a global overturning check need be performed.

Unfactored soil-concrete friction coefficients for sliding verifications shall be assumed equal to 0.45, and 0.50 under blast/impact loads conditions.

Passive soil resistance may be considered in sliding resistance calculation on a case by case basis provided there are no nearby underground cable trenches and shall be subject to the approval of the COMPANY; where included, the resistance factor on the passive soil resistance shall be taken as 0.5.

With regards to the overburden load in bearing capacity assessments, the full overburden until finished grade will be relied upon and the scenario of future removal of overburden will not be considered.

A vibration analysis shall be included in case the foundation is subjected to vibrations (e.g. induced by equipment and/or piping).

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4.2.2 Piled foundations

Piled foundations are not included in the onshore scope.

4.2.3 Foundation protection

Where required, the foundations shall be protected with two layers of bituminous emulsion.

The coating shall be applied to major foundations, process liquid retaining pits, RC box culverts for process piping, RC culverts for road crossing, RC bridge foundations and major RC retaining walls.

Major foundations include pipe rack foundations, equipment structure foundations, building foundations and equipment foundations identified in plot plan.

The coating is not required for concrete of drainage components such as manholes, pits, reinforced concrete pipes, reinforced concrete encasement for U/G pipe and ditches.

The coating is not required for minor foundations such as pipe sleepers, pipe supports, minor RC retaining walls, operating stage/walkway foundations, stair/ladder foundations, electrical/instrument panel foundations, lighting pole foundations, lightning pole foundations, CCTV foundations, fence and gate foundations and miscellaneous foundations.

The coating is not required for concrete road, paving areas and slab on grade. The coating is not required for lean concrete, cable duct banks, cable trenches and cable protection tiles/cable markers.

4.3 EXCHANGERS AND HORIZONTAL VESSELS

Tube bundle extraction force and wind loads shall be calculated according to specification T-14.111.002 "Design Basis for Loads and Load Combinations".

The horizontal loads, acting along the equipment longitudinal axis, shall be applied to the fixed saddle only.

In case of stacked exchanges, the higher bundle section shall be considered for the bundle extraction.

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Saddle design shall be carried out for the more loaded saddle (fixed saddle). Concrete gross section and reinforcement shall be the same for sliding and fixed saddle.

The sliding plate anchors shall be verified for transversal loads (wind, etc.).

4.4 **BUILDING AND STRUCTURES**

Reinforced concrete buildings and structures shall be designed in accordance with ACI 318M, considering the design loads and combinations defined in specification T-14.111.002 "Design Basis for Loads and Load Combinations".

4.5 **RETAINING STRUCTURES**

Reinforced concrete retaining structures (i.e. sumps, basins, etc.) shall be designed according to ACI 350M.

Liquid retaining structures shall be designed for the relevant load combinations in the following conditions:

- Erection/maintenance: Empty basin, earth pressure and ground water pressure acting on the outside of the walls, other loads as applicable (e.g. traffic loads on the ground outside the basin).
- Test: Basin full of water, disregarding the external soil pressure assuming that the surrounding soil is not present during hydrostatic test condition.
- Operation: Basin with liquid at operating level, earth pressure and ground water pressure acting on the outside of the walls, other loads as applicable (e.g. traffic loads on the ground outside the basin)

The crack width calculations for liquid retaining structures shall be determined in accordance with the limit state principles and formulae presented within Eurocode 2, considering a crack width limit = 0.1 mm in accordance with ACI 224R.

The use of several small-diameter bars with reduced spacing instead of few large-diameter bars widely spaced is recommended to limit the crack width.

The horizontal reinforcement in walls shall be generally placed closer to the

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concrete surface, on the outer side respect to the vertical bars.

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4.6 FOUNDATIONS OF VIBRATING MACHINERY

Light machinery foundations are defined as follows:

- foundations supporting rotating equipment including small centrifugal and vertical pumps, engines, turbines and compressors having reciprocating or rotary masses, with a total weight less than 2268 kg or a gross plan area of less than 2.5 m², centrifugal machinery less than 375 kW or reciprocating machinery less than 150 kW.

For the light machinery the ratio of foundation mass to equipment mass for rotary and reciprocating foundations shall be a minimum of 3:1 and 5:1 respectively. For the light machinery the dynamic analysis is not required.

Heavy machinery foundations are excluded from this document and shall be designed in accordance with GFS IG.34.85.25-Gen. "Heavy Machinery Foundations for Onshore LNG Facilities" and Amendment sheet (T-14.111.018).

5 DESIGN CRITERIA FOR CONCRETE PAVING

Concrete paving shall be designed in accordance with AASHTO Guide for Design of Pavement Structures, considering the loads defined in project specification T-14.111.002 "Design Basis for Loads and Load Combinations".

Support of miscellaneous vertical load of 11 kN or less, in paved areas may be made integral with the slab on grade.

Paving may be used a foundation base for the loading conditions, pedestal design and paving types as per the following table:

Baseplate loads on the pedestal	Pedestal dimensions, dead weight and paving type
N _{max} =11kN M _{max} =0.6kNm T _{max} =0.6kNm	Maximum size 400mm x 400mm x 400mm, 1.6kN Founded on <u>light duty paving</u>
N _{max} =11kN M _{max} =2.1kNm T _{max} =2.0kNm	Maximum size 400mm x 400mm x 400mm, 1.6kN Founded on <u>heavy duty paving</u>
Baseplate loads on the pedestal	Pedestal dimensions, dead weight and paving

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Nmax=11kN Mmax=0.7kNm Tmax=1.5kNm	Maximum size 500mm x 500mm x 400mm, 2.5kN Founded on <u>light duty paving</u>
Nmax=11kN Mmax=2.7kNm Tmax=6.0kNm	Maximum size 500mm x 500mm x 400mm, 2.5kN Founded on <u>heavy duty paving</u>
Baseplate loads on the pedestal	Pedestal dimensions, dead weight and paving type
Nmax=11kN Mmax=0.9kNm Tmax=1.9kNm	Maximum size 650mm x 650mm x 400mm, 4.2kN Founded on <u>light duty paving</u>
Nmax=11kN Mmax=6.0kNm Tmax=11.5kNm	Maximum size 650mm x 650mm x 400mm, 4.2kN Founded on <u>heavy duty paving</u>

Notes

Any type of paving load shall be separated by a minimum of 750mm distance from the edge of these pedestals.

The minimum distance between supports, centreline to centreline, applicable for heavy duty paving and for light duty paving, is presented in the table below:

plate dim. [mm]	Pedestal dim	Min distance between support
200x200mm	400x400x400mm	750mm
250x250mm	400x400x400mm	750mm
300x300mm	500x500x400mm	850mm
350x350mm	500x500x400mm	850mm
400x400mm	650x650x400mm	1000mm
450x450mm	650x650x400mm	1000mm

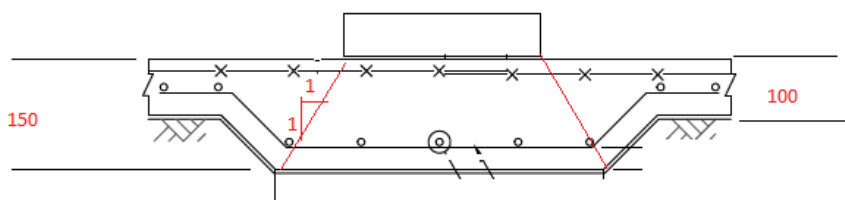
Alternative distances shall be subject to demonstration of feasibility.

The minimum distance between these pedestals and paving expansion joint shall be 500mm.

In case of supports with bending moments or shear loads exceeding the above listed thresholds for light duty paving, the loads may be distributed by locally

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thickening of the paving up to 150 mm as shown in the figure below; in such case the loading threshold will be as for heavy duty paving.



5.1 TYPES OF PAVING

Concrete paving shall be classified as follows (refer to T-14.112.006):

- Light duty paving: minimum thickness 100 mm
single welded wire mesh $\phi 7$ / 200x200 mm
- Heavy duty paving: minimum thickness 150 mm
double welded wire mesh $\phi 7$ / 200x200 mm

Fibre-Reinforced Concrete (FRC) may be alternatively used in accordance with ACI 544.1R.

5.2 JOINTS

Concrete paved areas shall be provided with:

- Expansion joints (ref. par. 5.2.1) to be provided along the perimeter of each panel and around foundations and pits. Expansion joints shall be located not more than 20 m apart.
- Contraction joints (ref. par. 5.2.2) to be located between expansion joints, spaced not more than 8 m apart.
- Construction joints (ref. par. 5.2.3) shall be carried out at each interruption of concrete pouring.

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5.2.1 Expansion joints

Expansion joints shall be 20 mm wide and shall extend to the full depth of the concrete paving slab. Joint shall extend through curbs.

The reinforcement shall not extend across the joint.

If required, for high duty paving, dowel bars may be provided across the expansion joints.

5.2.2 Contraction joints

Contraction joints shall be realized creating a 10 mm wide x 20 mm deep continuous groove at the top surface of the concrete slab.

The reinforcement shall be continuous across the joint.

5.2.3 Construction joints

Construction joints shall be provided where the concreting operations are interrupted for a period sufficient for the concrete to harden.

The joint surface of the hardened concrete shall be chipped, washed and wetted to saturation before pouring fresh concrete.

Construction joints shall be perpendicular to the main reinforcement and located away from points of peak demand so as to least impair the performance of the structure.

The reinforcement shall be continuous across the joint.

5.2.4 Joint sealant

Sealer shall be chemical resistant (regarding possible spills) two-component polysulfide compound. Sealer for down hand application shall be liquid self-levelling grade in accordance with ASTM D6690. Sealer for vertical joints or for overhead application shall be trowel grade and in accordance with ASTM C881/C881M Type IV, Grade 3, Class B/C.

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6 ATTACHMENTS

T-14.111.000-SH2

Evaluation Of Rigid Behavior Of Vertical Vessel Foundations

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